

CONTINUOUS INTEGRATION VS CONTINUOUS DEPLOYMENT



What is Continuous Integration (CI)?

- **Frequent Code Merges:**

Developers push code often to a shared repository.

- **Automated Builds & Tests:**

Each commit triggers automated build and test pipelines.

- **Early Bug Detection:**

Helps catch integration issues early in the development cycle.

- **Popular Tools:**

Jenkins, GitHub Actions, GitLab CI.

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What is Continuous Deployment (CD)?

- **Auto Deploy to Production:**

Every successful CI build is automatically released.

- **No Manual Intervention:**

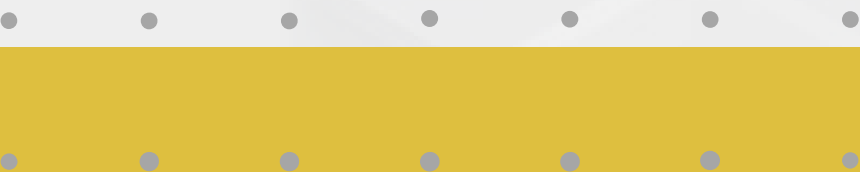
Eliminates the need for manual approval or deployment steps.

- **Fast Feature Delivery:**

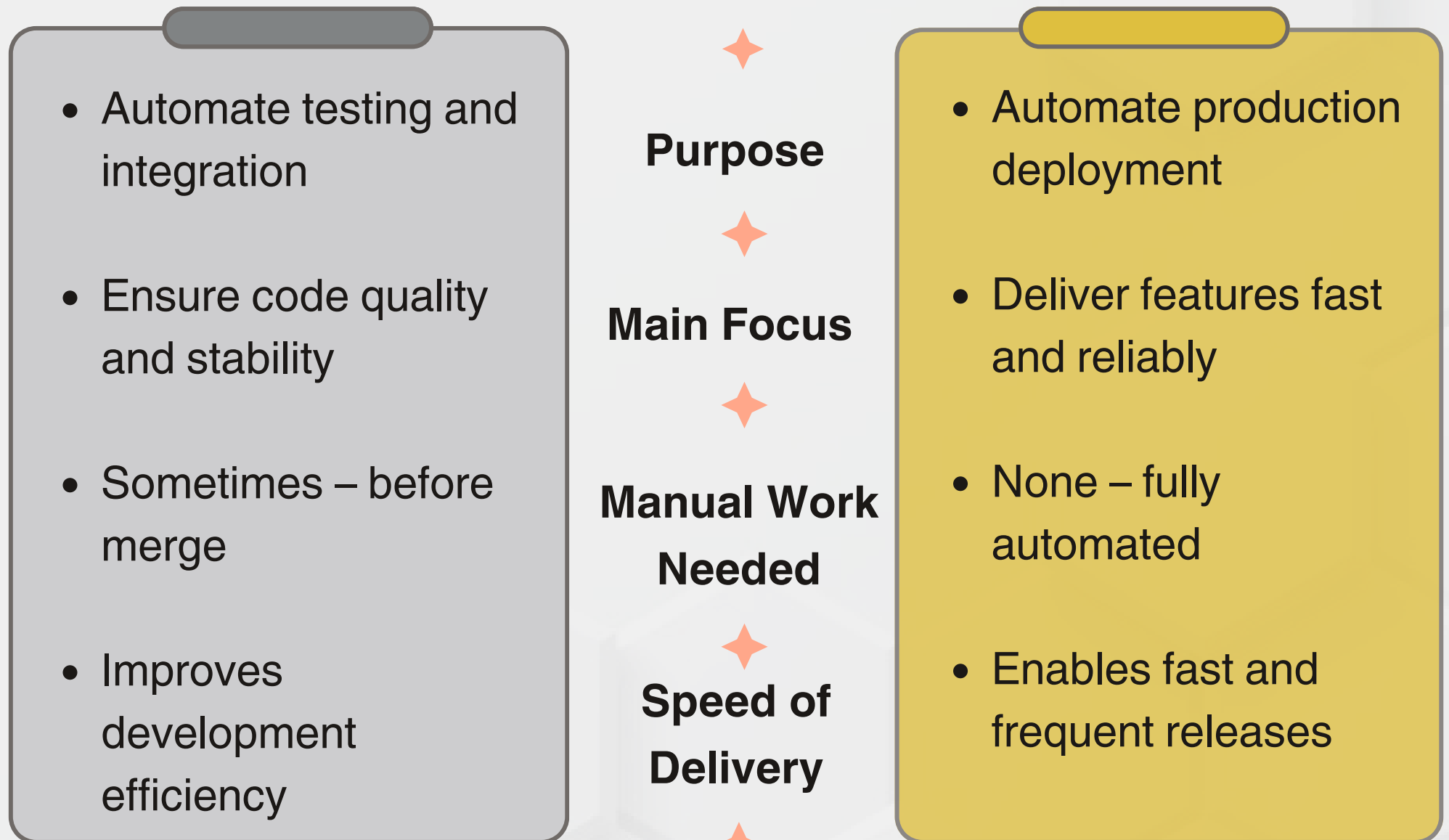
Enables rapid rollout of updates, features, and fixes.

- **Popular Tools:**

ArgoCD, Spinnaker, GitLab CD.



CI vs CD – Key Differences



Why Use CI/CD Together?

- **Reliable Pipelines:**

Reduce errors with tested, repeatable workflows.

- **Shorter Time to Market:**

Ship features faster and stay competitive.

- **Greater Team Confidence:**

Developers focus on innovation, not deployment stress.

